

## Education

2020-24 **Bioengineering Research PhD, Imperial College London**

Supervisor: Dario Farina

2016-20 **MEng Biomedical Engineering, Imperial College London**

First Class Honours (78.14)

Year 3 Dean's List, Best Poster award

## Research Experience

2020-24 **PhD:** Learned shared-autonomy locomotion control using RL for human and intent-driven robotic prosthesis over challenging terrain in simulated environments I built. Collected my own experimental motion and biosignal datasets and processed them with TCNs to extract high-level user intent. **1<sup>st</sup> place presentation in category at iCBEI22, 2 first-author articles published in IEEE journals, 2 more under review.**

2016-20 **MEng:** Projects included parallelised image processing algorithms with CNNs for brain-cell counting, building accessible UI for controlling various assistive technology. Built my own data loading, systematic training and documentation systems for my thesis project; EMG based kinematics prediction and activity classifier. **Projects presented at AAATE18, BioMedEng19, BiomedEng22 conferences.**

2024/25 Coauthor of the **NeurIPS** competition MyoChallenge, developed simulated scenes for RL in humanoid control tasks. **Accepted in 2025 to D&B track.**

## Work Experience

2025 - **ICL/Meta:** Postdoctoral researcher at the Wearable Neural Interfaces Research Centre. Developed neuromechanical models of the hand, and control policies using RL. Built fully forward muscle signal synthesis pipeline in massively parallelized simulated environments for data augmentation. Planning and writing for multiple grant applications.

2023-25 **Google DeepMind:** Contractor to develop features for the Unity plugin for DeepMind's MuJoCo physics engine and provide community moderation and answers. Managed issues and fixed bugs for other related projects too.

2023 **Artanim Foundation:** Paid research internship position. Investigated social spacing in virtual environments. Developed tools and processes for physics-based animation. Implemented interactive environments in VR. Planned and performed full-body, multi-actor motion capture sessions.

## Skills

- **IT:** Python, Jax, ML frameworks (TF Keras, PyTorch, Flax), Unity, C#, Arduino, Blender, Git, WPF, C++
- Released and documented [an open-source RL humanoid control toolkit](#)
- **Theory skills:** Signal processing, Supervised and Reinforcement Learning, Information theory
- **Soft Skills:** Coordinated teaching assistance on university course. Designed and led multi-day workshops for simulating wearable robotics. Organized student conference. Supervised and guided students on vision-based robotics control, mocap suits, sensor simulation and estimator uncertainty modelling.
- **Writing & Research:** Performed systematic reviews, provided peer-review to IEEE journals, [Publications listed on Google Scholar](#)